ANSWER KEY — Integration Lesson 5 Test: The Power Rule for Antiderivatives

For the grader · full working shown · a missing + C on a family answer costs half; equivalent forms (e.g. $(1/3)x^3$ vs $x^3/3$) are fine

1. Find the antiderivative of x^3.

> **Work:** raise the power by one (x^4), divide by the new power 4. Check: derivative of $x^4/4 = 4x^3/4 = x^3$ ✓. **Answer: $x^4/4 + C$.**

2. Find the antiderivative of 4 (just the constant 4).

> **Work:** a constant k has antiderivative kx (a line with slope k). Check: derivative of $4x + C = 4$ ✓. **Answer: $4x + C$.**

3. Find the antiderivative of $9x^2$.

> **Work:** the 9 rides along: $9 \cdot x^3/3 = 3x^3$. Check: derivative of $3x^3 = 9x^2$ ✓. **Answer: $3x^3 + C$.** ($9x^3/3 + C$ unsimplified is acceptable but ask for simplification.)

4. Find the antiderivative of $x^4 + x^2$.

> **Work:** term by term: $x^4 \to x^5/5$; $x^2 \to x^3/3$. One + C covers the sum. Check: $5x^4/5 + 3x^2/3 = x^4 + x^2$ ✓. **Answer: $x^5/5 + x^3/3 + C$.**

5. Find the antiderivative of $6x^5 + 2x$.

> **Work:** $6x^5 \to 6 \cdot x^6/6 = x^6$; $2x \to 2 \cdot x^2/2 = x^2$. Check: derivative of $x^6 + x^2 = 6x^5 + 2x$ ✓. **Answer: $x^6 + x^2 + C$.**

6. Find the antiderivative of $x + 7$.

> **Work:** $x \to x^2/2$; $7 \to 7x$. Check: $2x/2 + 7 = x + 7$ ✓. **Answer: $x^2/2 + 7x + C$.**

7. Check whether $x^6/6 + x$ is an antiderivative of $x^5 + 1$.

> **Work:** differentiate the candidate: derivative of $x^6/6 = 6x^5/6 = x^5$; derivative of $x = 1$. Total $= x^5 + 1$ — exact match with the target. **Answer: yes — its derivative is exactly $x^5 + 1$, so it is an antiderivative.** (The written differentiation must appear for full credit.)

8. A student says the antiderivative of x^2 is $x^3 + C$. Check and fix if needed.

> **Work:** check: derivative of $x^3 + C = 3x^2$, NOT x^2 — three times too big. The student raised the power but forgot to divide by the new power. Fix: $x^3/3 + C$. Check the fix: $3x^2/3 = x^2$ ✓. **Answer: wrong — the correct antiderivative is $x^3/3 + C$ (must divide by the new power 3).**

9. Find the antiderivative of $(x + 1)(x + 4)$.

> **Work:** multiply out first: $(x + 1)(x + 4) = x^2 + 4x + x + 4 = x^2 + 5x + 4$. Then term by term: $x^2 \to x^3/3$; $5x \to 5x^2/2$; $4 \to 4x$. Check: derivative $= x^2 + 5x + 4$ ✓. **Answer: $x^3/3 + 5x^2/2 + 4x + C$.**

10. Find $f(x)$ given $f'(x) = 6x^2 - 2$ and $f(1) = 5$, then verify both facts.

> **Work:** antiderivative: $6x^2 \to 6 \cdot x^3/3 = 2x^3$; $-2 \to -2x$. So $f(x) = 2x^3 - 2x + C$. Point: $f(1) = 2 - 2 + C = 0 + C = 5 \to C = 5$. Verify: derivative of $2x^3 - 2x + 5 = 6x^2 - 2$ ✓; $f(1) = 2 - 2 + 5 = 5$ ✓. **Answer: $f(x) = 2x^3 - 2x + 5$.**